

MATH 213 – Homework 7

The following two questions are about the textbook data set called “acs12”. The data set is a relatively small (2000-case) sample drawn from the American Community Survey (ACS) from the year 2012. This survey is an ongoing project conducted by the US Census Bureau based on very carefully constructed (representative) samples of US residents. It provides researchers and policymakers a wide range of data about people living in the US. Our textbook authors took a random sample of 2000 cases from the 2012 ACS data set (which has millions of cases). Because of the way that the ACS is designed, you can think of this sample as essentially a random sample drawn from all US residents in 2012.

You will find the codebook for the acs12 data set at

<https://www.openintro.org/data/index.php?data=acs12>

As always, write in complete sentences, justifying your answers appropriately.

1. Use the codebook and what I wrote about the ACS above to answer the following general questions about the study.

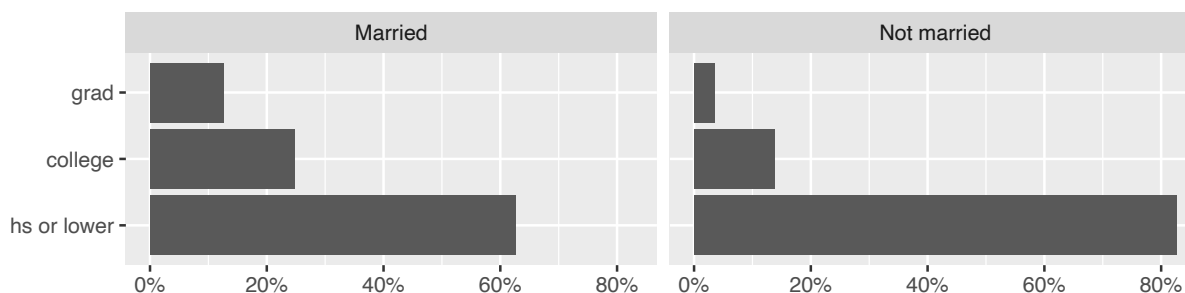
- a. **Our textbook authors have provided us with a sample (based on the ACS) to examine. What is the population of interest?**

The population of interest is all U.S. residents in 2012.

- b. **Is it reasonable to generalize whatever we learn from the sample to this larger population? Explain.**

Yes, it is reasonable to generalize findings from this sample to the larger US population. Because it was randomly drawn meaning it has a high chance of reflecting broader population additionally a sample size of 2000 is relatively large and should be a reliable way to collect data.

2. **The following chart shows the distribution of the “education level” variable in the ACS data set, grouped by whether the person was married.**



Write a few sentences describing the differences you see in education levels between married and non-married people in the sample. Start with your who/what/where/when sentence. Then, launch

into your description by starting with the big point you would like to make, and support it using numbers you draw from the graphs.

The data comes from the 2012 ACS survey the graphs above compare the distribution of education level (grad, college and HS or lower) of individuals who are married or are not. As we can see married individuals generally have higher education levels while individuals who are not married are more concentrated in the lower education levels where over 80 percent fall into the HS or lower category. In contrast those who are married only make up roughly 60 percent for that same category and the remaining are located in the higher education levels. This pattern might suggest an association between education and marriage rates.